

A large graph-feature dictionary, one column at a time

2026

Abstract

The attribution is univariate. Each graph feature is one column, scored by the same concept-drift SSE drop and the same mean shift. A dictionary of ten million columns is therefore the same procedure as the dictionary of twenty-three columns already run. On this machine a screen of 2×10^4 noise columns costs $14.8 \mu\text{s}$ each, which extrapolates to about 150 seconds for 10^7 columns. Eighty permutations on every one of those columns is a few hours, so the permutations belong on the columns the screen ranks first. The measured searches show why the dictionary can be large: most columns stay silent, and the outcome drop sits on a few of them.

1 The column

An episode is one finished search. $Y \in \{0, 1\}$ is whether that search returned a solved state. A graph feature is any number attached to the committed path: depth, children, siblings, a centrality, a path-score moment, a state coordinate. Write x for one such number across n episodes, and write batch for the indicator of episodes after the frozen reference. The concept contribution of that column is

$$\Delta(x) = \text{SSE}(Y \sim 1 + \text{batch}) - \text{SSE}(Y \sim 1 + \text{batch} + x + \text{batch} \cdot x).$$

The covariate contribution is the absolute change in the mean of x across the drift. Neither statistic looks at any other column. A dictionary $x_1, \dots, x_p$ is p repetitions of this pair. No joint model of the ten million columns is required to rank them.

2 What a small dictionary already did

Two searches, forty episodes, drift at episode 16. The dictionary had twenty-three columns in six layers. Shares are sums of positive $\Delta(x)$ inside the layer.

Layer	Game of 24	One-step offer
Statistics	0.354	0.000
History	0.294	0.732
Interaction	0.164	0.136
State encoding, four numbers	0.095	0.132
Graph	0.094	0.000
Context	0.000	0.000

On Game of 24, Y falls from 1.00 to 0.25. Score entropy, the rolling mean, and score times path length each have permutation $p = 0.012$. Sibling count, the best graph column, moves by 0.88 and

has $p = 0.074$. Beam width and maximum depth contribute $\Delta = 0$. On the one-step offer search the tree shape is constant, so every graph column has $\Delta = 0$ and mean shift 0. Y falls from 0.50 to 0.167. The previous outcome takes share 0.732, with $p = 0.012$. The offer-identity coordinate shifts by 1 and has concept $p = 0.42$: it restates the batch bit, and the interaction does not add a separate piece of Y .

Those zeros are the reason to admit a large dictionary. Columns that do not move, and columns that only rename the batch, fall out. The ranking does not need a weight assigned to “graph” in advance.

3 Cost of ten million columns

The screen below is one least-squares fit of four coefficients per column, on $n = 40$, with no permutation. Twenty thousand standard-normal columns took 0.296 seconds on this machine, $14.8\mu\text{s}$ per column. At that rate, 10^7 columns take about 150 seconds. The same microbenchmark with 80 permutations on 50 columns cost 2.3 ms per column, which extrapolates to a few hours for every one of 10^7 columns. The permutations were counted with two fits inside the shuffle loop, so the figure is an upper envelope rather than a tuned kernel.

The practical order is the one the statistic suggests. Screen all p columns by $\Delta(x)$. Permute the top of that list, or permute until the p -values separate. A million copies of one step function will tie with each other and with the batch; they do not become a million discoveries. Sibling count is a discovery only if its own Δ and its own shuffle still stand after the copies are added.

4 What is not in the dictionary

This run did not build 10^7 graph features, and it did not fit a 768-dimensional encoder. The state encoding is four numbers. The current episode’s residual is not a column, because that residual contains Y . Unit prices and AUC coefficients are not inputs. The claim that does follow from the runs is narrower: the attribution is a sort of columns, the sort is cheap relative to opening a budget on a tied column, and a fixed tree contributes a block of exact zeros that a larger graph dictionary is free to contain.